# The Deep Body

Exploring users' insides through design fiction

Harun Kaygan*
Department for Design, Media and Educational Science, University of Southern Denmark, Kolding, Denmark
hkay@sdu.dk

Çağlar Genç
Gamification Group, Tampere University, Tampere, Finland
genc.caglar@tuni.fi

Oğuz 'Oz' Buruk
Gamification Group, Tampere University, Tampere, Finland
oguz.buruk@tuni.fi

Ida Kathrine Hammeleff Jørgensen
Department for Design, Media and Educational Science, University of Southern Denmark, Kolding, Denmark
idka@sdu.dk

Linas Kristupas Gabrielaitis
Gamification Group, Tampere University, Tampere, Finland
linas.gabrielaitis@tuni.fi

## ABSTRACT

Design approaches in the last two decades have shown extensive interest in and developed diverse strategies for taking users' bodily capabilities and embodied experiences into account both within the design process and in the design outcomes. This workshop focuses on one specific way in which users' bodies have been characterized and addressed by design practice across such diverse approaches as wearable technologies, design for self-tracking, somadesign, etc.: 'the deep body.' In this context, 'the deep body' refers to the various manners in which design activities engage with users' bodies with the consideration that they possess an internal dimension, whether literal or metaphorical, that can be explored through embodied design. Based on this definition, the workshop will actively engage participants in a collaborative exploration of the now and future of 'deep bodies' in design. We will utilize presentations, discussions and embodied and speculative ideation and so build a collection of fictional abstracts that outline design approaches and outcomes, and practice-oriented definitions for considering the 'depth' of the body.

## CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); HCI design and evaluation methods; User models; Human computer interaction (HCI); HCI theory, concepts and models; Interaction design; Interaction design theory, concepts and paradigms.

## KEYWORDS

body-centred design, embodied design, speculative design, fictional abstracts

**ACM Reference Format:**
Harun Kaygan, Çağlar Genç, Oğuz 'Oz' Buruk, Ida Kathrine Hammeleff Jørgensen, and Linas Kristupas Gabrielaitis. 2024. The Deep Body: Exploring users' insides through design fiction. In *Proceedings of the 27th International Academic Mindtrek Conference (Mindtrek '24), October 08–11, 2024, Tampere, Finland*. ACM, New York, NY, USA, 5 pages. https://doi.org/10.1145/3681716.3681743

## 1 BACKGROUND

### 1.1 Embodied users and deep bodies in HCI research

Somatic turn [34] in the field of human-computer interaction (HCI) from the late 1990s onwards has meant intensified interest in and engagement with embodied experiences of future users. Early engagements, e.g. in [39] and [13] soon led the way to a wide range of manners in which users are conceived in embodied terms, including but not limited to considerations of full-body, expressive movement [15, 25, 53], skilled practice [12, 29] and embodied sense making [26, 30]. Based in phenomenological and somaesthetic traditions, among others, scholars argued that designers need to develop vocabularies, tools as well as embodied sensibilities to appreciate and address users' lived, situated, embodied experiences [22, 33, 42, 48]. Such work has been often contrasted with "solutionist" approaches to HCI that rely overly on sensor-based (self-)tracking for solving health problems or self-knowledge and enhancement [2, 21, 24].

Reviewing these developments across HCI, Homewood et al. [21] identify a series of moments, from the introduction of concerns with embodiment to HCI, to the diversification of such concerns, more recently on to more-than-human approaches to the body. In another review paper, Kaygan et al. [28] investigate embodied user representations across design and design-led HCI literatures, and so draw attention to the expanding views as to what aspects of the body can and should be addressed by design activity. According to

*Corresponding author


*Mindtrek '24, October 08–11, 2024, Tampere, Finland*

ACM ISBN 979-8-4007-1823-6/24/10
https://doi.org/10.1145/3681716.3681743

the authors, one significant aspect of that expansion has been a concern with the "internal," either conceived literally with reference to internal bodily processes, or figuratively by evoking internal senses such as proprioception as well as somatic awareness, achieved e.g. through meditation.

Here we call "the deep body" those conceptions of the user whereby designers imagine and address users' bodies as having an internal dimension, be it literal or figurative, that can be accessed through embodied design practices.

In many cases, the insides refer to bodily processes that are often inaccessible by regular senses. HCI research, including studies of bodily experience, has made widespread use of biosignals [1], such as heart [41] and breath rates [47]. Research through design examples have utilized tracking of as well as creative engagement with bodily fluids such as menstruation blood [9, 16], breastmilk [19] and saliva [20]. The concept of inbodied interaction has been proposed as a way to involve HCI field capitalize more effectively on the complexity and plasticity of physiological systems, with references from designing for circadian rhythms and eating habits [44].

In other cases, the "insides" refer to internal senses, also formulated as the "feel dimension:" how one experiences the world from the first-person point-of-view through proprioception and spatial awareness, but oriented towards the world as action potentials [31, 48]. This sense of somatic experience has been further expanded, particularly by inspiration from somaesthetics, in order that design research can engage with embodied, affective states and felt experiences. Somadesign [22] applies practices such as meditation, Feldenkraus and Alexander-technique [23], to dehabituate embodied experience, turn attention inwards, reflect and foster a deep understanding and aesthetic appreciation of "inner experiences" of the body and its parts, processes, functions and movements, such as breathing [52], bowel function [45], muscle movements in the pelvic area [46]. Somadesign thus cultivates somatic connoisseurship [23], that is trains designers' as well as users' aesthetic sensibilities so as to let them achieve a non-habituated, deeper and more nuanced experience and understanding of their own body.

### 1.2 Design fiction as a means to explore deep bodies

Speculative and Critical Design, along with Design Fiction, have primarily been employed to adopt an anti-solutionist approach in HCI [14]. Speculative design seeks to challenge prevailing cultural and societal norms by crafting artifacts that embody the contexts, ideas, ideologies, and narratives of future societies [14]. Critical design, on the other hand, diverges from speculative design by deliberately engaging with future scenarios in a critical manner [14]. Design Fiction, a central tool in speculative and critical design practices, encompasses the practices of world-building, character creation, design, and speculation [5]. It has garnered increasing attention from HCI researchers through the creation of fictional abstracts [11], user studies [5], fictional probes/prototypes [37], and even fictional papers [32]. Design Fiction enables the depiction of worlds influenced by technological artifacts, incorporating stakeholders often overlooked, such as non-human actors or marginalized communities. It offers a balanced perspective on the future, presenting both utopian and dystopian scenarios [51], or what is termed as "ustopias" [36]. While Design Fiction involves narratives of unknown futures, its primary goal is to provoke reflection on the present, prompting sense-making, action, and paradigm shifts [6]. For instance, a Design Fiction narrative about habitation on Mars may spark reflections on the necessity for such endeavors and how to address environmental transformations caused by climate change. Similarly, fictional abstracts can shed light on critical perspectives regarding the potential social consequences and experiences resulting from the integration of technology into bodies and minds [8].

Capable of questioning current and future implications of practices, we argue that design fiction provides a fruitful approach for exploring the established and emergent qualities of specific ways user are imagined, addressed and engaged with in design – in this case, 'deep bodies' – that are found widespread across design explorations in the field of HCI. As such, design fiction will be used in this workshop to elaborate on the underlying assumptions and extrapolate design practical as well as societal implications of a specific albeit varied conception of users.

## 2 WORKSHOP GOALS & THEMES

This workshop aims to bring together practitioners and researchers to discuss now(s) and future(s) of deep bodies in design. By discussing our current approaches and speculating futures about how bodies are conceptualized, utilized and intervened in design, we aim to critically engage with the 'depth' of how bodies are considered in design practices. Accordingly, in the workshop, we will consider, but not be limited to, the following themes as lenses to examine deep bodies of design:

### 2.1 Defining 'Bodies' and 'Depth'

Examining a realm in design where the conceptions of bodies are fluent and everchanging [21], in this theme, we will critically engage with the ways that we define 'bodies' and 'depth.' We will discuss for instance: What are the boundaries of a 'body,' e.g. molecular definition of bodyhood, homeostasis and plasticity, more-than-human approaches that indicate human bodies' entanglement with both microbial life and nonhuman animals, etc.? What metaphors do we use to understand the depth of bodies, e.g. surface/depth and inside/outside? What does it mean to look 'inside' oneself and/or others in techno-centric, biomedical, psychological, empathic, etc. ways? What do we expect to see when we look 'inside'?

### 2.2 Dealing with Depth through Designs

Design and design research has focused on exploring the design outcomes that might enable a deeper understanding of bodies. We can identify some trends in that direction: For example, **(1) design as excavation** where designers look under the skin for innovation potentials, showing interest in biological knowledge (e.g. use of biosensors, inbodied design [44] etc.); **(2) design as translation**, where designers render all kinds of bodily experiences visible and/or tangible, trackable and open to manipulation and experience (e.g. [17, 18]); **(3) design as guidance**, where designers guide and aid users towards better understanding, identification, management and repair with an eye to personal objectives, challenges and concerns (e.g. through tracking [10, 40], guidance [47], sharing [38] etc.). By discussing and extending these approaches, in this theme, we will speculate different ways design outcomes can question and extrapolate from existing understandings as to what a deep understanding of bodies means and entails.

### 2.3 Methods for Understanding Deep Bodies in Design

Researchers are moving away from viewing bodies solely as objects of design and are instead creating methods that put bodies at the forefront of the design process, aiming for deeper involvement and comprehension of bodies in design. These methods include activities like using their bodies to sketch ideas [35], and immersing in design contexts through role-playing, exercises that heighten bodily awareness (i.e., sensitizing [50]) and observations from both first and third-person perspectives [49], as well as utilizing body maps to better attune to their bodies [3]. Our workshop will critically reflect on state-of-the-art methods and speculate new ones to enhance our understanding of deep bodies in design.

## 3 WORKSHOP STRUCTURE

This one-day workshop will be conducted in person, allowing individuals to not only present and discuss their visions of 'deep bodies' in design, but also engage in roleplaying activities in which they will be able to enact and reflect on each other's visions. Two weeks before the workshop, we will prompt participants to submit a brief presentation of their vision related, but not limited, to workshop themes. The workshop will be planned as an all-day, 8-hour-long activity. It will consist of five parts:

- **Show & Tell** (2h): Participants will be invited to briefly present and reflect on their visions. This will enable participants to acknowledge the diversity of views toward deep bodies of design by contrasting and discussing each other's works.
- **Embodied Ideation** (1.5h): Participants will work as a group and select one of the works presented in the previous session. They will use embodied ideation activities (i.e., bodystorming [43], embodied sketching [35]) to create speculative designs based on the selected works by using papers and other simple materials (i.e., experience prototypes, paper prototypes). These ideas will create a basis for the role-playing activities in the following sessions.
- **Lunch** (1h)
- **Role-playing & Presentations** (2h): The groups will use ideas to engage in a role-playing activity to present the visions in an embodied manner. They will be asked to imagine characters to act as and contexts to role-play in to explore and present what kind of worlds, experiences and societal impacts the ideas could bring in. During the process of planning their role-playing presentations, they will be supported with templates for creating characters, and roughly defining a context for their role-playing presentations. After approximately an hour of preparations, each group will make a presentation in which they role-play as characters within the visions.
- **Reflective Discussions & Writing Fictional Abstracts** (2h): Building on the prior sessions of presenting, prototyping and enacting the visions for deep bodies in design, we will have a discussion session where we reflect on opportunities and challenges of understanding deep bodies in design. This discussion will aim to combine and contrast approaches, as well as outline critiques regarding the now and future of deep bodies in design. This discussion will also focus on outlining several Fictional Research Abstracts [32] based on the points discussed and the visions presented. For this, the organizers will create a template to help participants write the abstracts. Finally, we will discuss further steps (i.e., how to turn the visions & fictional abstracts into a publication and how to stay in touch for further discussions and collaborations).

## 4 INTENDED OUTCOMES AND DISSEMINATION PLANS

Given our aim of gathering individuals interested in critically engaging with the 'depth' of how bodies are considered in design practices, we intend to use this workshop as a starting point for unpacking now(s) and future(s) of deep bodies in design. One of the outcomes we envision this community to make is a collection of fictional abstracts that map out critical and speculative methods, designs and definitions for 'depth' of body in design research. We believe that such a contribution could fit follow-up joint publications, e.g. an extended abstract in CHI about fictional visions for 'deep bodies' (similar to [4, 8]) and a full paper in IMWUT about defining, designing for and methods of addressing deep bodies in design. Second, we intend to use the workshop as a starting point for a larger, possibly ongoing conversation around exploring deep bodies in design. To achieve this, we will establish a Discord channel with the participants for future collaborations and discussions. To attract a wider audience, we will also put together and distribute a visual poster that summarizes the outcomes of the workshop through social media channels and a workshop website.

## 5 CALL FOR PARTICIPATION

This one-day workshop invites researchers and practitioners to discuss how bodies are conceptualized, utilized, and engaged with in design as things with depth. Throughout a mixture of presentations, discussions, and embodied and speculative design activities, we aim to critically engage with the 'depth' of how bodies are considered in design practices.

We invite candidates to submit a 2-6-page pictorial or paper (i.e., position papers, case studies, design fictions, speculative and critical designs). The submissions are expected to present a current or a future vision about how the view toward bodies in design are or could be deepened in relation to one or more of the three workshop themes: **(1) Defining 'Bodies' and 'Depth':** What is a body and what does it mean to have a 'deep' understanding of bodies in design? **(2) Dealing with Depth through Designs:** In what ways do we and can we achieve a "deep" understanding of bodies through design? What can we design to do so? **(3) Methods**

**for Understanding Deep Bodies in Design:** What kind of design methods are there, could there be to understand and address bodies in their depth? The submission should be in PDF and submitted via e-mail (**hkay@sdu.dk**). They will be reviewed based on relevance and contribution to the workshop.

## 6 ORGANIZERS

**Harun Kaygan** is Associate Professor of Design Culture at the University of Southern Denmark, Kolding. He is interested in the cultural and political role and implications of design practices and outcomes. His current research concerns how design engages with users' bodies and well-being today.

**Çağlar Genç** is a postdoctoral researcher in the Gamification Group at Tampere University. His current research focuses on the intersection of MtH and interaction design. He explores human augmentation technologies to investigate interaction and communication opportunities among humans and non-human entities.

**Oğuz 'Oz' Buruk** is an Assistant Professor of Gameful Experience at Tampere University, Finland. His research focuses on designing gameful environments for various contexts such as body-integrated technologies, computational fashion, posthumanism, urban spaces, extended reality and nature. He frequently employs methods such as speculative design, design fiction and participatory design.

**Ida Jørgensen** is a postdoc at the University of Southern Denmark, Kolding. Her research focuses on how the (gendered) body is imagined, modeled and enacted in design of games and playable artifacts and technologies.

**Linas Kristupas Gabrielaitis** is a doctoral researcher at the Gamification Group, Tampere University, Finland. His research focuses on games and gaming, map sketching and diagramming as speculative/ artistic practices for more-than-human design.

## ACKNOWLEDGMENTS

The research is funded by the Carlsberg Foundation, grant number CF21-0231.

We have used ChatGPT 3.5 in the Abstract and Section 2 as a rephraser (see [7]).

## REFERENCES

[1] Alfaras, M., Tsaknaki, V., Sanches, P., Windlin, C., Umair, M., Sas, C. and Höök, K. 2020. From Biodata to Somadata. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu HI USA, Apr. 2020), 1–14.

[2] Almeida, T., Balaam, M., Bardzell, S. and Hansen, L.K. 2020. Introduction to the Special Issue on HCI and the Body: Reimagining Women's Health. *ACM Transactions on Computer-Human Interaction.* 27, 4 (Sep. 2020), 1–32. DOI:https://doi.org/10.1145/3406091.

[3] Anne Cochrane, K., Mah, K., Ståhl, A., Núñez-Pacheco, C., Balaam, M., Ahmadpour, N. and Loke, L. 2022. Body Maps: A Generative Tool for Soma-based Design. *Sixteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Daejeon, Republic of Korea, Feb. 2022), 1–14.

[4] Baumer, E.P.S. *et al.* 2014. CHI 2039: speculative research visions. *CHI '14 Extended Abstracts on Human Factors in Computing Systems* (Toronto Ontario Canada, Apr. 2014), 761–770.

[5] Baumer, E.P.S., Blythe, M. and Tanenbaum, T.J. 2020. Evaluating Design Fiction: The Right Tool for the Job. *Proceedings of the 2020 ACM Designing Interactive Systems Conference* (Eindhoven Netherlands, Jul. 2020), 1901–1913.

[6] Blythe, M. 2014. Research through design fiction: narrative in real and imaginary abstracts. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto Ontario Canada, Apr. 2014), 703–712.

[7] Buruk, O. "Oz" 2023. Academic Writing with GPT-3.5 (ChatGPT): Reflections on Practices, Efficacy and Transparency. *26th International Academic Mindtrek Conference* (Tampere Finland, Oct. 2023), 144–153.

[8] Buruk, O. "Oz" *et al.* 2020. Children in 2077: Designing Children's Technologies in the Age of Transhumanism. *Extended Abstracts of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu HI USA, Apr. 2020), 1–14.

[9] Campo Woytuk, N., Nilsson, L. and Liu, M. 2019. Your Period Rules: Design Implications for Period-Positive Technologies. *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow Scotland Uk, May 2019), 1–6.

[10] Coşkun, A. and Karahanoğlu, A. 2023. Data Sensemaking in Self-Tracking: Towards a New Generation of Self-Tracking Tools. *International Journal of Human–Computer Interaction.* 39, 12 (Jul. 2023), 2339–2360. DOI:https://doi.org/10.1080/10447318.2022.2075637.

[11] Coulton, P., Lindley, J., Sturdee, M. and Stead, M. 2017. Design Fiction as World Building. *Research through Design Conference.* (2017), 4230036 Bytes. DOI:https://doi.org/10.6084/M9.FIGSHARE.4746964.

[12] Djajadiningrat, T., Matthews, B. and Stienstra, M. 2007. Easy doesn't do it: skill and expression in tangible aesthetics. *Personal and Ubiquitous Computing.* 11, 8 (Oct. 2007), 657–676. DOI:https://doi.org/10.1007/s00779-006-0137-9.

[13] Dourish, P. 2001. Where the Action is: The Foundations of Embodied Interaction. MIT Press.

[14] Dunne, A. and Raby, F. 2013. Speculative everything: design, fiction, and social dreaming. The MIT Press.

[15] Fogtmann, M.H., Fritsch, J. and Kortbek, K.J. 2008. Kinesthetic interaction: revealing the bodily potential in interaction design. *Proceedings of the 20th Australasian Conference on Computer-Human Interaction Designing for Habitus and Habitat - OZCHI '08* (Cairns, Australia, 2008), 89.

[16] Fox, S., Howell, N., Wong, R. and Spektor, F. 2019. Vivewell: Speculating Near-Future Menstrual Tracking through Current Data Practices. *Proceedings of the 2019 on Designing Interactive Systems Conference* (San Diego CA USA, Jun. 2019), 541–552.

[17] Fyhn, C. and Buur, J. 2019. A Tangible Understanding of Chronic Pain. Proceedings of the NORDES Conference: Who Cares? (Helsinki, Finland, Jun. 2019).

[18] Gomez Ortega, A., Bourgeois, J. and Kortuem, G. 2022. Reconstructing Intimate Contexts through Data Donation: A Case Study in Menstrual Tracking Technologies. *Nordic Human-Computer Interaction Conference* (Aarhus Denmark, Oct. 2022), 1–12.

[19] Helms, K. 2021. Entangled Reflections on Designing with Leaky Breastfeeding Bodies. *Designing Interactive Systems Conference 2021* (Virtual Event, USA, Jun. 2021), 1998–2012.

[20] Homewood, S., Bewley, H. and Boer, L. 2019. Ovum: Designing for Fertility Tracking as a Shared and Domestic Experience. *Proceedings of the 2019 on Designing Interactive Systems Conference* (San Diego CA USA, Jun. 2019), 553–565.

[21] Homewood, S., Hedemyr, M., Fagerberg Ranten, M. and Kozel, S. 2021. Tracing Conceptions of the Body in HCI: From User to More-Than-Human. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama Japan, May 2021), 1–12.

[22] Höök, K. 2018. Designing with the body: somaesthetic interaction design. The MIT Press.

[23] Höök, K. *et al.* 2018. Embracing First-Person Perspectives in Soma-Based Design. *Informatics.* 5, 1 (Feb. 2018), 8. DOI:https://doi.org/10.3390/informatics5010008.

[24] Höök, K., Eriksson, S., Louise Juul Søndergaard, M., Ciolfi Felice, M., Campo Woytuk, N., Kilic Afsar, O., Tsaknaki, V. and Ståhl, A. 2019. Soma Design and Politics of the Body. *Proceedings of the Halfway to the Future Symposium 2019* (Nottingham, United Kingdom, Nov. 2019), 1–8.

[25] Hummels, C., Overbeeke, K.C. and Klooster, S. 2007. Move to get moved: a search for methods, tools and knowledge to design for expressive and rich movement-based interaction. *Personal and Ubiquitous Computing.* 11, 8 (Dec. 2007), 677–690. DOI:https://doi.org/10.1007/s00779-006-0135-y.

[26] Hummels, C. and Van Dijk, J. 2015. Seven Principles to Design for Embodied Sensemaking. *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction* (Stanford California USA, Jan. 2015), 21–28.

[27] Jørgensen, I.K.H. and Kaygan, H. 2024. The Body in Play: Dimensions of Embodiment in Design for Play. *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction* (Cork Ireland, Feb. 2024), 1–12.

[28] Kaygan, H., Jørgensen, I.K.H. and Karpova, J. In review. With a Body for a Body: Mapping Design's Imaginaries of Human Bodies. *Design Issues.* (In review).

[29] Kilbourn, K. and Buur, J. 2007. The patient as skilled practitioner. *Nordes 2007: Design Inquiries* (Stockholm, Sweden, 2007).

[30] Klemmer, S.R., Hartmann, B. and Takayama, L. 2006. How bodies matter: five themes for interaction design. *Proceedings of the 6th ACM conference on Designing Interactive systems - DIS '06* (University Park, PA, USA, 2006), 140.

[31] Larssen, A.T., Robertson, T. and Edwards, J. 2007. The feel dimension of technology interaction: exploring tangibles through movement and touch. *Proceedings of the 1st international conference on Tangible and embedded interaction - TEI '07* (Baton Rouge, Louisiana, 2007), 271.

[32] Lindley, J. and Coulton, P. 2016. Pushing the Limits of Design Fiction: The Case For Fictional Research Papers. *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose California USA, May 2016), 4032–4043.

[33] Loke, L. and Robertson, T. 2011. The lived body in design: mapping the terrain. *Proceedings of the 23rd Australian Computer-Human Interaction Conference* (Canberra Australia, Nov. 2011), 181–184.

[34] Loke, L. and Schiphorst, T. 2018. The somatic turn in human-computer interaction. *Interactions*. 25, 5 (Aug. 2018), 54–5863. DOI:https://doi.org/10.1145/3236675.

[35] Márquez Segura, E., Turmo Vidal, L., Rostami, A. and Waern, A. 2016. Embodied Sketching. *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose California USA, May 2016), 6014–6027.

[36] Noortman, R., Funk, M., Andersen, K. and Eggen, B. 2021. What Would Margaret Atwood Do? Designing for Ustopia in HCI. *Academic Mindtrek 2021* (Tampere/Virtual Finland, Jun. 2021), 72–80.

[37] Noortman, R., Schulte, B.F., Marshall, P., Bakker, S. and Cox, A.L. 2019. HawkEye - Deploying a Design Fiction Probe. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow Scotland Uk, May 2019), 1–14.

[38] Núñez-Pacheco, C. and Loke, L. 2022. Focusing for Interaction Design: An Introspective Somatic Method. *CHI Conference on Human Factors in Computing Systems* (New Orleans LA USA, Apr. 2022), 1–18.

[39] Robertson, T. 1997. Cooperative Work and Lived Cognition: A Taxonomy of Embodied Actions. *Proceedings of the Fifth European Conference on Computer Supported Cooperative Work*. Springer Netherlands. 205–220.

[40] Sanches, P., Janson, A., Karpashevich, P., Nadal, C., Qu, C., Daudén Roquet, C., Umair, M., Windlin, C., Doherty, G., Höök, K. and Sas, C. 2019. HCI and Affective Health: Taking stock of a decade of studies and charting future research directions. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow Scotland UK, May 2019), 1–17.

[41] Schiphorst, T. 2009. Body Matters: The Palpability of Invisible Computing. *Leonardo*. 42, 3 (Jun. 2009), 225–230. DOI:https://doi.org/10.1162/leon.2009.42.3.225.

[42] Schiphorst, T. 2011. Self-evidence: applying somatic connoisseurship to experience design. *CHI '11 Extended Abstracts on Human Factors in Computing Systems* (Vancouver BC Canada, May 2011), 145–160.

[43] Schleicher, D., Jones, P. and Kachur, O. 2010. Bodystorming as embodied designing. *Interactions*. 17, 6 (Nov. 2010), 47–51. DOI:https://doi.org/10.1145/1865245.1865256.

[44] Schraefel, M.C. 2020. Inbodied Interaction: Introduction. *Interactions*. 27, 2 (Feb. 2020), 32–37. DOI:https://doi.org/10.1145/3380811.

[45] Sommer, L.K., Lauritzen, J.M., Spork, A., Biller, L., Jensen, M.M. and Bemman, B. 2024. I've Gut Something to Tell You: A Speculative Biofeedback Wearable Art Installation on the Gut-Brain Connection. *ArtsIT, Interactivity and Game Creation*. A.L. Brooks, ed. Springer Nature Switzerland. 305–324.

[46] Ståhl, A., Balaam, M., Ciolfi Felice, M. and Kaklopoulou, I. 2022. An Annotated Soma Design Process of the Pelvic Chair. *Designing Interactive Systems Conference* (Virtual Event Australia, Jun. 2022), 1921–1933.

[47] Ståhl, A., Jonsson, M., Mercurio, J., Karlsson, A., Höök, K. and Banka Johnson, E.-C. 2016. The Soma Mat and Breathing Light. *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems* (San Jose, CA, USA, May 2016), 305–308.

[48] Svanæs, D. 2013. Interaction design for and with *the lived body*: Some implications of Merleau-Ponty's phenomenology. *ACM Transactions on Computer-Human Interaction*. 20, 1 (Mar. 2013), 1–30. DOI:https://doi.org/10.1145/2442106.2442114.

[49] Svanæs, D. and Barkhuus, L. 2020. The Designer's Body as Resource in Design: Exploring Combinations of Point-of-view and Tense. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu HI USA, Apr. 2020), 1–13.

[50] Tajadura-Jimenez, A., Ley-Flores, J., Valdiviezo, O., Singh, A., Sanchez-Martin, M., Diaz Duran, J. and Márquez Segura, E. 2022. Exploring the Design Space for Body Transformation Wearables to Support Physical Activity through Sensitizing and Bodystorming. *Proceedings of the 8th International Conference on Movement and Computing* (Chicago IL USA, Jun. 2022), 1–9.

[51] Thibault, M., Buruk, O. "Oz," Buruk, S.S. and Hamari, J. 2020. Transurbanism: Smart Cities for Transhumans. *Proceedings of the 2020 ACM Designing Interactive Systems Conference* (Eindhoven Netherlands, Jul. 2020), 1915–1928.

[52] Tsaknaki, V., Balaam, M., Ståhl, A., Sanches, P., Windlin, C., Karpashevich, P. and Höök, K. 2019. Teaching Soma Design. *Proceedings of the 2019 on Designing Interactive Systems Conference* (San Diego CA USA, Jun. 2019), 1237–1249.

[53] Wilde, D., Schiphorst, T. and Klooster, S. 2011. Move to design/design to move: a conversation about designing for the body. *Interactions*. 18, 4 (Jul. 2011), 22–27. DOI:https://doi.org/10.1145/1978822.1978828.